

TSRMTGen: Leveraging LLMs for Financial Narrative Summaries from Risk Models

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Abstract. Financial risk models provide critical quantitative insights, yet translating complex risk data into actionable narratives remains a challenge. While LLMs offer powerful text generation capabilities, they struggle with numerical reasoning and are prone to hallucinations which is a critical flaw in financial compliance environment. This paper proposes TSRMTGen, a framework that transforms risk model data into accurate, coherent narratives. Our approach integrates expert knowledge through "Hint" injection and utilizes a two-step generation process to ensure factual consistency. We demonstrate the effectiveness of TSRMTGen through a rigorous 3-layer human-in-the-loop evaluation, achieving a 0% error rate in generating summaries for investment portfolios.

Keywords: risk model · LLM · text generation · narrative summaries

1 Introduction & Related Work

Large language models (LLMs) have been utilized to assist educational learning [11, 12], conversations [18, 17], simulations [17], outlier detection [3, 10], multi-objective optimization [16, 6], etc. In finance, LLMs can transform complex data and visualizations into clear, actionable insights, such as in portfolio management [2, 8, 14, 15, 13, 18, 5]. Risk models provide detailed quantitative outputs and visualizations [4, 9], but users often struggle to interpret them. LLMs could generate summaries from risk model data, yet challenges remain: they lack numerical reasoning, require domain knowledge, and must comply with strict regulations that allow no factual errors [7, 1]. This paper leverages LLMs to interpret risk model data and generate narrative summaries from them. To our knowledge, no prior work has addressed this. Key challenges include:

- Complexity and domain expertise: Risk models involve nonlinear equations, stochastic processes, and domain-specific assumptions that LLMs may misrepresent.
- Pattern handling: LLMs struggle to extract useful patterns from numerical data.
- Hallucination and factual errors: LLMs may produce plausible but inaccurate narratives, yet the financial industry demands zero factual errors [7, 1].

We propose the Two-Step Risk Model Text Generation (TSRMTGen) framework, combining expert knowledge with LLMs to produce coherent narratives. TSRMTGen

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enhances interpretability, improves user comprehension, and supports informed decision-making. Its effectiveness is validated through a human-in-the-loop evaluation, marking the first attempt to generate error-free narrative explanations from financial risk data.

2 Methodology: The TSRMTGen Framework

The workflow of the proposed TSRMTGen framework can be described by Figure 1. More specifically, there are three major components – data interpretation, pattern discovery with expert knowledge, and two-step LLM text generation.

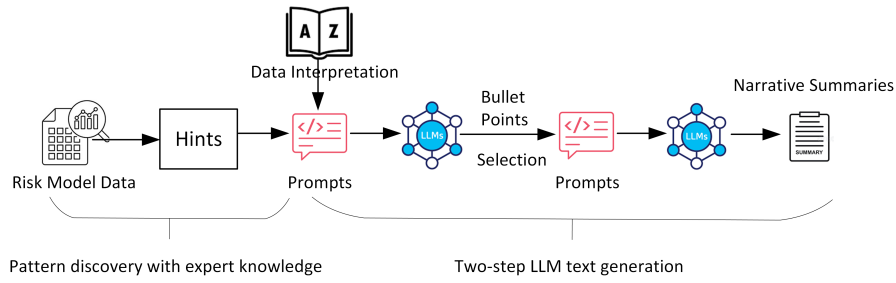


Fig. 1: Workflow of the TSRMTGen Framework

Data Interpretation. Most LLMs are trained on large-scale texts in public and may possess a general understanding of financial knowledge. However, they lack a deep comprehension of the financial numbers or terms from a specific risk model. In our studies, we adopted the risk model proposed by Morningstar, Inc. which focuses on 7 key factors, including size, style, quality, yield, liquidity, volatility and momentum of a given financial portfolio. The outputs of these key factors were utilized as the inputs for LLMs. To aid the LLM, we inject data interpretations into the prompt context, where these interpretations are pre-defined expert knowledge to indicate the impacts of each factor in the risk model. For example, we may explicitly define that "a positive relative exposure to size indicates a bias towards larger companies compared to the benchmark". This contextual grounding reduces ambiguity of the financial rules or patterns.

Pattern Discovery with Expert Knowledge. Directly asking an LLM to interpret complex time-series plots often leads to errors. We solve this by pre-processing data into *Expert Hints*. A "Hint" is a deterministic rule that translates numerical conditions into texts. These hints were categorized into two types. The first category is *conditional hints* which trigger specific descriptions based on thresholds. For example, financial experts may define a threshold on factor contributions to determine if a factor had a positive or negative impact on returns based on the threshold. These hints are used to derive financial patterns when specific conditions are satisfied. The second category is *inferential hints* which explain cause-and-effect relationships. For example, combining exposure bias with factor premiums to explain why a portfolio gained or lost value

(e.g., "Overweight in volatility contributed to loss because high-volatility stocks underperformed"). We have defined 4 inferential hints and 33 conditional hints based on expert knowledge in order to better infer financial patterns. We also calculate an "Interesting Score" (0-5) for each factor based on its impact magnitude. Only factors with high scores are selected for the final summary, ensuring the narrative focuses on the most critical risks.

Two-Step LLM Text Generation. To eliminate hallucinations, we decouple reasoning from writing. We generate bullet points in the first step. More specifically, we prompt the LLM to generate structured bullet points for each risk factor. The prompt includes the hints generated in the previous stage. This constrains the LLM to predefined logical boundaries. The LLM produces specific JSON outputs for the key factors as well as the interesting scores. In the second step, we generate financial narrative summaries. We feed the selected bullet points (filtered by Interesting Score ≥ 2) into the LLM. The model's task is strictly to synthesize these points into a coherent, professional narrative (approx. 400 words). By forcing the model to rely on the verified bullet points, we prevent it from inventing data or drawing unauthorized conclusions.

3 Evaluation and Results

We employed financial experts to manually review the generated narrative summaries and identify the errors in the output texts from LLM. These financial experts are coming from fund research and quant research team and have profound knowledge about risk model data. They will identify three types of errors – factual errors (i.e., wrong numbers cited), inference errors (i.e., wrong logic) and style errors (i.e., informal languages).

Given a large number of potential fund portfolios (more than 250,000 in our database), we propose and employ a 3-layer iterative evaluation process, as depicted by Figure 2.

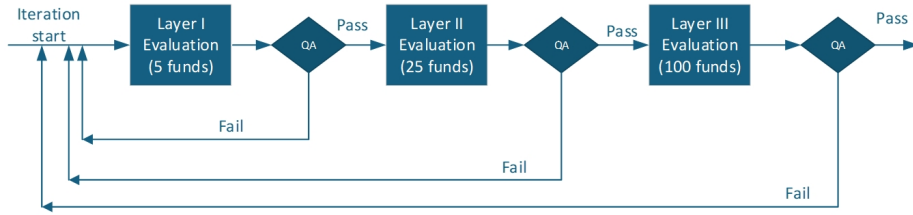


Fig. 2: 3-Layer Iterative Evaluation Process

- **Layer I with 5 standard mutual funds.** At the beginning, we selected 5 representative mutual funds and evaluate the summaries generated for them. We tune the components in our framework until no errors appeared.
- **Layer II with 25 random selected funds.** In Layer II, we implement a human-in-the-loop refinement process. The summaries are evaluated, and the system components are iteratively adjusted to ensure the outputs are error-free. In ten iterations,

we randomly evaluated 25 mutual funds per iteration, with two financial experts conducting Layer II QA assessments to ensure accuracy and reliability.

- **Layer III with 100 random selected funds.** We evaluate the TSRMTGen framework by using 100 random selected mutual funds. 7 experts were employed for evaluations, stopping only if no errors were found in 100 summaries; Layer III tests robustness over a larger portfolio, again with expert QA.

As depicted by Figure 2, as long as we have identified errors, we need to go back and start from the initial stage, i.e., Layer I, since we need to ensure that the revisions will not bring errors in Layer I and can fix errors in other layers.

Considering output quality, cost, and efficiency, we selected Claude 3.5 Sonnet as our LLM. Table 1 shows accuracy, the number of mutual funds with errors, specific errors per iteration, and the number of prompt revisions to improve subsequent iterations.

Most revisions involved structure changes to ensure correct formatting, such as avoiding ‘%’ for percentiles. ‘Conditional hints’ revisions updated rules defined by experts. ‘Inferential hints’ revisions refined cause-and-effect relationships requiring expert knowledge.

From iteration 7, accuracy improved significantly, and by iteration 9 no errors were identified. Iteration 10 confirmed this. In Layer III, 100 randomly selected mutual funds were evaluated with no errors, demonstrating TSRMTGen’s robustness. The framework is now integrated into our financial dashboard, enhancing transparency by providing accurate summaries and explanations to clients.

Table 1: Iterative Evaluations and Refinements in Layer II

Iteration #	Accuracy	# of Funds with Errors	Number of Identified Errors			Number of Revisions After Current Iteration			
			Factual Errors	Inference Errors	Style Errors	Data Interpretation	Conditional Hints	Inferential Hints	Prompts
1	72%	7	1	5	1	1	1	1	2
2	67%	8	3	8	0	0	1	2	1
3	62%	10	4	6	1	0	1	1	1
4	71%	7	0	6	1	0	2	1	3
5	67%	8	2	8	0	0	1	3	4
6	71%	7	0	5	2	0	1	2	3
7	95%	1	1	0	0	0	1	1	1
8	92%	2	0	2	0	0	1	1	0
9	100%	0	0	0	0	0	0	0	0
10	100%	0	0	0	0	0	0	0	0

4 Conclusions and Future Work

We present TSRMTGen, a framework that generates narrative summaries of financial portfolios by combining risk-model data with expert knowledge and LLMs, bridging complex datasets and clear explanations. Future work includes developing quantitative text-quality metrics and A/B testing dashboard enhancements with the generated summaries.

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